

ROBOT DATA

Episode / session

one demonstration; episodes recorded consecutively at one site

End-effector

the tool at the arm's tip, a two-finger gripper

Action block

all commands of one model step, concatenated

SHIFTWM TERMS

Patch features

frozen DINOv2 16×16 grid

Transport

content moved from observed patches

Gate

per-patch weight: keep vs. move

Transport window

w×w patches in the last S frames

Correction

added residual for new content

Oracle move

observed feature closest to the truth

SURGICAL TERMS

dVRK

da Vinci Research Kit, a surgical robot

Scene camera

an external camera, not an endoscope

Language-Table

xArm pushing blocks; long unscripted play data

DROID

real Franka arm episodes; split by recording session

SIMULATED SUITES

PushT

push a T-shaped block to a target pose

Wall

2-D agent moves through a door in a wall

TwoRoom

2-D navigation between two rooms (LeWM)

Reacher

a two-link arm reaches a target (LeWM)

MPC / CEM

plan, act, re-plan; sample and refit

EVALUATION

Persistence

copy the last observed grid

Moving patches

25% with the largest true change

Skill

share of persistence error removed

Rollout

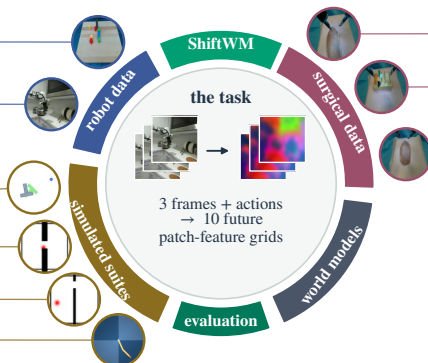
own predictions fed back as inputs

Teacher forcing

train on true past, not own output

Session split

no recording session in two splits



• **Open-H Hamlyn**
dVRK recordings of seven practice tasks

• **Practice task**
a standard surgeon exercise, e.g. peg transfer

• **Phantom**
an artificial practice model of tissue

WORLD MODELS

V-JEPA 2-AC
published action-conditioned WM, post-trained on DROID

DINO-WM
published WM on DINOv2 patch features

LeWM
released end-to-end WM; planning reference

Direct
regresses each future grid as Z_0 + residual

AR
residual to the latest grid, rolled out recursively

AR-TF
DINO-WM-style: teacher-forced next step, rolled out